

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2023

Course: B. Tech.

Semester :III

Branch: Electrical Engineering / Electrical Engineering (Electronics and Power)/

Electrical & Electronics Engg. / Electrical & Power Engineering.

Subject Code & Name: BTES305 Engineering material science

Max Marks: 60

Date:11-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define the following terms	CO1	6
A) Drift velocity B) Mobility C) Collision time D) Mean free path E) Relaxation time		
B) Provide a detailed explanation of Joule's Law with a neat diagram.	CO2	6
C) What is Electron Scattering? And what are the deferent types of Electron Scattering.	CO2	6
Q.2 Solve Any Two of the following.		12
A) Give an explanation of the structure of crystalline and amorphous crystals using a suitable diagram and example.	CO1	6
B) What is dielectric constant and dielectric strength of material? And what are the significant of dielectric constant and dielectric strength?	CO3	6
C) What is the definition of polarization in dielectric materials and what is its classification?	CO3	6
Q. 3 Solve Any Two of the following.		12
A) What is Semiconductor? Write its properties in detail.	CO3	6
B) Explain intrinsic semiconductor and extrinsic semiconductor with suitable examples	CO3	6
C) Provide an explanation of the Einstein relationship in semiconductor materials.	CO2	6
Q.4 Solve Any Two of the following.		12
A) Provide appropriate diagrams and examples to explain the classification of magnetic materials.	CO2	6

B)	Explain Ferro magnetic material and write down properties of ferromagnetic material.	CO2	6
C)	What is curie temperature? And what are the significance of curie temperature/curie point in magnetic material?	CO1	6
Q. 5	Solve Any Two of the following.		12
A)	What is radioactivity, and how do radioactive materials differ from non-radioactive ones?	CO2	6
B)	Explain Bragg's law and its significance in X-ray crystallography. How does it relate to the interference of X-rays in a crystal lattice?	CO3	6
C)	What is ultra-sonic radiography? explain in in detail.	CO2	6

***** End *****